

ANALYTICAL PROBLEM-SOLVING ACTIVITY

ANSWERED

Topic 1. Metrics for Performance Evaluation

1. Accuracy

A classification model tested 100 samples. It correctly classified 85 samples.

1. How many samples were classified incorrectly?

Answer: Incorrect samples = $100 - 85 = 15$ samples.

2. Calculate the accuracy of the model.

Answer: Accuracy = (Correct predictions / Total predictions) $\times 100 = (85/100) \times 100 = 85\%$.

3. Calculate the error rate.

Answer: Error rate = (Incorrect predictions / Total predictions) $\times 100 = (15/100) \times 100 = 15\%$.

4. Is the model performing well based on accuracy?

Answer: Yes. The model has an accuracy of 85%, meaning it correctly classifies 85 out of 100 samples. This indicates reasonably good performance.

5. What happens to accuracy if 5 more samples are classified correctly?

Answer: Correct predictions become 90 out of 100. New accuracy = $(90/100) \times 100 = 90\%$. Therefore, accuracy increases from 85% to 90%.

2. Confusion Matrix

TP = 40, TN = 35, FP = 10, FN = 15.

1. What is the total number of predictions?

Answer: Total = TP + TN + FP + FN = $40 + 35 + 10 + 15 = 100$ predictions.

2. Calculate the accuracy.

Answer: Accuracy = (TP + TN) / Total = $(40 + 35) / 100 = 75\%$.

3. Calculate the error rate.

Answer: Error rate = (FP + FN) / Total = $(10 + 15) / 100 = 25\%$.

4. Which value represents incorrectly predicted positive cases?

Answer: False Positive (FP) = 10.

5. Which value represents incorrectly predicted negative cases?

Answer: False Negative (FN) = 15.

3. Precision and Recall

TP = 80, FP = 20, FN = 10, TN = 90.

1. Calculate Precision.

Answer: Precision = $TP / (TP + FP) = 80 / (80 + 20) = 80/100 = 0.80 = 80\%$.

2. Calculate Recall.

Answer: Recall = $TP / (TP + FN) = 80 / (80 + 10) = 80/90 = 0.8889 \approx 88.89\%$.

3. Which metric tells us how many predicted positive cases were actually positive?

Answer: Precision.

4. Which metric tells us how many actual positive cases were correctly identified?

Answer: Recall.

5. Which metric is more important when missing a positive case is dangerous?

Answer: Recall is more important because it measures the proportion of actual positive cases that are correctly identified. A high recall reduces false negatives.

4. Compare Two Classifiers

Metric	Classifier A	Classifier B
Accuracy	90%	85%
Precision	80%	95%
Recall	70%	90%

1. Which classifier has higher accuracy?

Answer: Classifier A, with 90% accuracy.

2. Which classifier has higher precision?

Answer: Classifier B, with 95% precision.

3. Which classifier has higher recall?

Answer: Classifier B, with 90% recall.

4. Which classifier would you choose if detecting as many positive cases as possible is important?

Answer: Classifier B, because it has the higher recall of 90%, so it correctly identifies more actual positive cases.

5. Give one reason why accuracy alone may not be enough.

Answer: Accuracy can be misleading when the classes are imbalanced. Precision and recall provide more specific information about positive-case performance.

5. Analyze Model Performance

TP = 70, TN = 80, FP = 20, FN = 30.

1. Calculate accuracy.

Answer: Total = 70 + 80 + 20 + 30 = 200. Accuracy = $(70 + 80) / 200 \times 100 = 75\%$.

2. Calculate precision.

Answer: Precision = $70 / (70 + 20) = 70/90 \approx 77.78\%$.

3. Calculate recall.

Answer: Recall = $70 / (70 + 30) = 70/100 = 70\%$.

4. Calculate the error rate.

Answer: Error rate = $(20 + 30) / 200 \times 100 = 25\%$.

5. Based on these values, identify one strength and one weakness of the model.

Answer: Strength: The model has reasonably good precision (77.78%), so most predicted positive cases are actually positive. Weakness: Recall is only 70%, meaning 30% of actual positive cases are missed.

Topic 2. Prediction Algorithms

1. Simple Linear Prediction

Advertising Cost: ₹10 → Sales 20; ₹20 → Sales 40; ₹30 → Sales 60; ₹40 → Sales 80.

1. Identify the relationship between advertising cost and sales.

Answer: There is a direct linear relationship. Sales increase by 20 units for every ₹10 increase in advertising cost. The relationship is $\text{Sales} = 2 \times \text{Advertising Cost}$.

2. Predict sales when advertising cost is ₹50.

Answer: Sales = $2 \times 50 = 100$ units.

3. What happens to sales when advertising cost increases by ₹10?

Answer: Sales increase by 20 units.

4. Is this an example of classification or prediction?

Answer: It is a numerical prediction (regression) problem.

5. Why is a linear prediction algorithm suitable here?

Answer: The given data follows a straight-line pattern, with sales increasing at a constant rate as advertising cost increases.

2. Predict Temperature

Given temperatures: Day 1 = 25°C, Day 2 = 26°C, Day 3 = 27°C, Day 4 = 28°C.

1. Predict the temperature on Day 5.

Answer: 29°C.

2. Predict the temperature on Day 10.

Answer: 34°C.

3. What is the predicted temperature on Day 15?

Answer: 39°C.

4. How much does the temperature increase for every additional day?

Answer: The temperature increases by 1°C per day.

5. Is this a classification or prediction problem?

Answer: It is a numerical prediction problem.

3. House Price Prediction

Area and price data: 1000 sq.ft → ₹40 lakhs; 1500 sq.ft → ₹60 lakhs; 2000 sq.ft → ₹80 lakhs.

1. Identify the relationship between area and price.

Answer: There is a direct linear relationship. Every additional 500 sq.ft increases the price by ₹20 lakhs. The price is ₹0.04 lakh per sq.ft.

2. Predict the price of a 2500 sq.ft house.

Answer: Price = $2500 \times 0.04 = ₹100$ lakhs.

3. What happens to the price when area increases by 500 sq.ft?

Answer: The price increases by ₹20 lakhs.

4. Which type of algorithm can be used for this problem?

Answer: Linear regression can be used.

5. Is this classification or numerical prediction?

Answer: It is numerical prediction (regression).

4. Compare Actual and Predicted Values

Actual Value	Predicted Value	Absolute Error
50	48	2
60	65	5
70	68	2
80	85	5
90	88	2

1. Which prediction has the smallest error?

Answer: Predictions of 48, 68, and 88 have the smallest absolute error of 2.

2. Which prediction has the largest error?

Answer: Predictions of 65 and 85 have the largest absolute error of 5.

3. Calculate the error for each prediction.

Answer: Absolute errors: $|50-48| = 2$, $|60-65| = 5$, $|70-68| = 2$, $|80-85| = 5$, and $|90-88| = 2$.

4. Is the model generally predicting close to the actual values?

Answer: Yes. All absolute errors are small, ranging from 2 to 5 units, so the predictions are generally close to the actual values.

5. Why is the difference between actual and predicted values important?

Answer: The difference, or prediction error, shows how accurately the model predicts the actual outcome. Smaller errors generally indicate better prediction performance.

5. Choosing a Prediction Algorithm

A. Predicting tomorrow's temperature. B. Predicting the price of a house. C. Predicting the number of products sold next month. D. Predicting whether an email is spam or not.

1. Which applications are prediction problems?

Answer: A, B, and C are numerical prediction problems.

2. Which application is a classification problem?

Answer: D is a classification problem because the output belongs to categories such as spam or not spam.

3. Which algorithm could be used for predicting house prices?

Answer: Linear regression could be used when the relationship between house features and price is approximately linear.

4. Why is regression suitable for numerical prediction?

Answer: Regression is designed to predict continuous numerical values, such as temperature, price, sales, or demand.

5. Give one difference between classification and prediction.

Answer: Classification predicts a discrete class or category, whereas numerical prediction (regression) predicts a numerical value.